

An anatomical illustration of the human brain and cranial nerves, rendered in a dark, textured style. The brain is shown from a superior view, with the cerebellum and brainstem visible at the bottom. Various cranial nerves are labeled with numbers 1 through 9. The text 'Brain & Cranial Nerves I' is overlaid in a large, white, serif font.

Brain & Cranial Nerves I

BIOL2251

Chapter 14

Section 14.1 – 14.8

Lecture Quiz



Test what you know!



8 questions (MC + **Matching**)



Quiz automatically submits at 2:30 PM



Password = brain

Make sure you sign in before the end of lecture!

Surface Landmarks



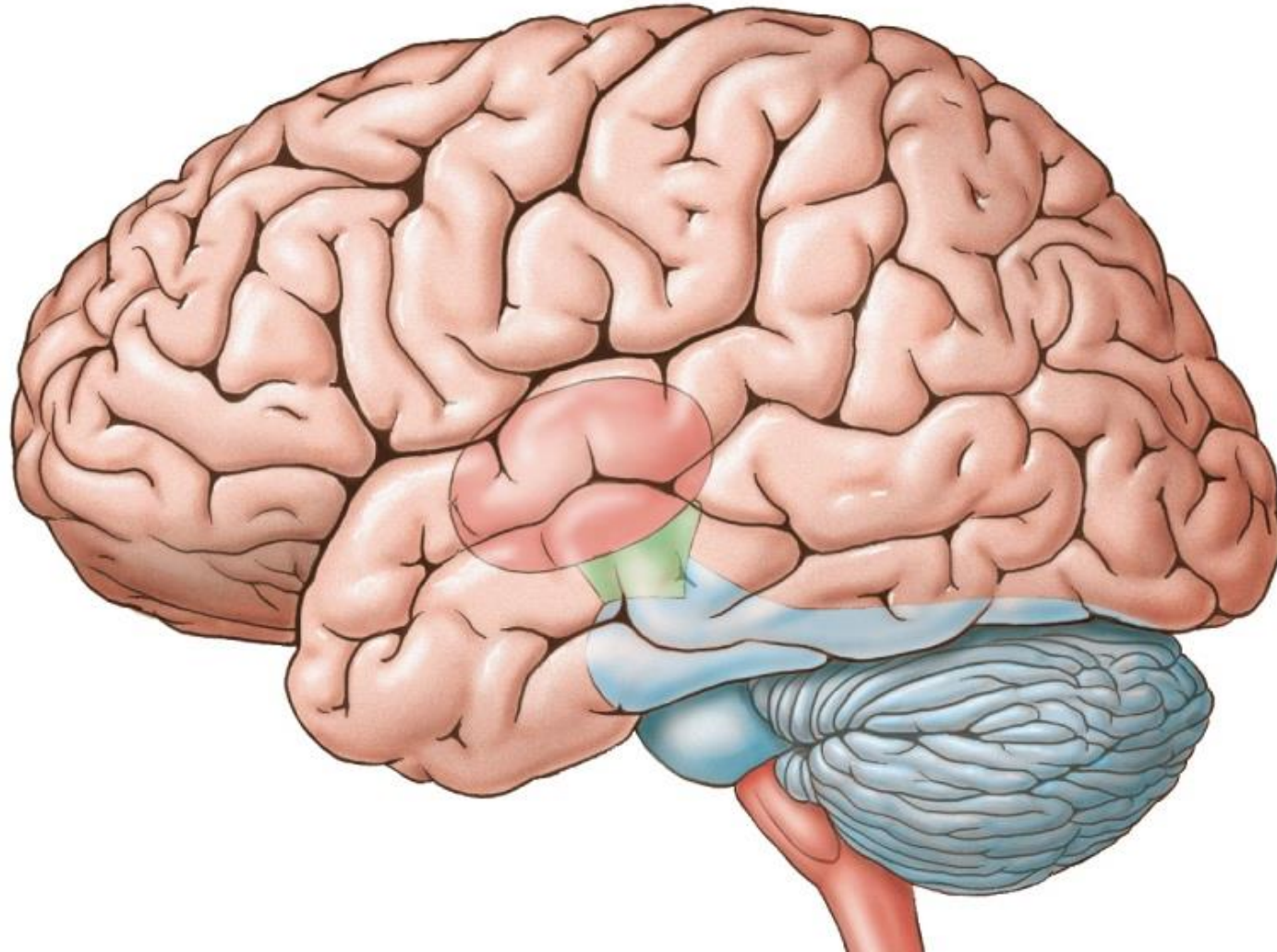
Surface Landmarks

- **Instructions**

- I will provide a blank, lateral view of the brain
- In pairs:
 - **Identify and label a gyrus, sulcus, and fissure**
 - **Explain to you partner why the brain is folded**
- I will ask for volunteers to identify structures and explain why the brain is folded

- **Bloom's taxonomy = Remember/Understand**

Surface Landmarks



Meninges Matching



Meninges Matching

- **Instructions**

- Work in small groups (~5 students)
- **Match the description to the correct structure within the meninges**
- We will review as a class
- **Bloom's taxonomy = Understand/Apply**

Meninges Matching

Term	Description
Dura mater	A. Directly adheres to the brain surface and follows its contours
Arachnoid mater	B. Contains CSF and blood vessels that cushion the brain
Pia mater	C. Dense outer layer; supports dural venous sinuses
Subarachnoid space	D. Web-like middle layer that bridges over sulci
Dural sinuses	E. Channels within dura that drain venous blood
Epidural space	F. Potential space between skull and dura (contains arteries in trauma)

Meninges Matching Key

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Dura mater	C. Dense outer layer; supports dural venous sinuses
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CSF Production & Circulation

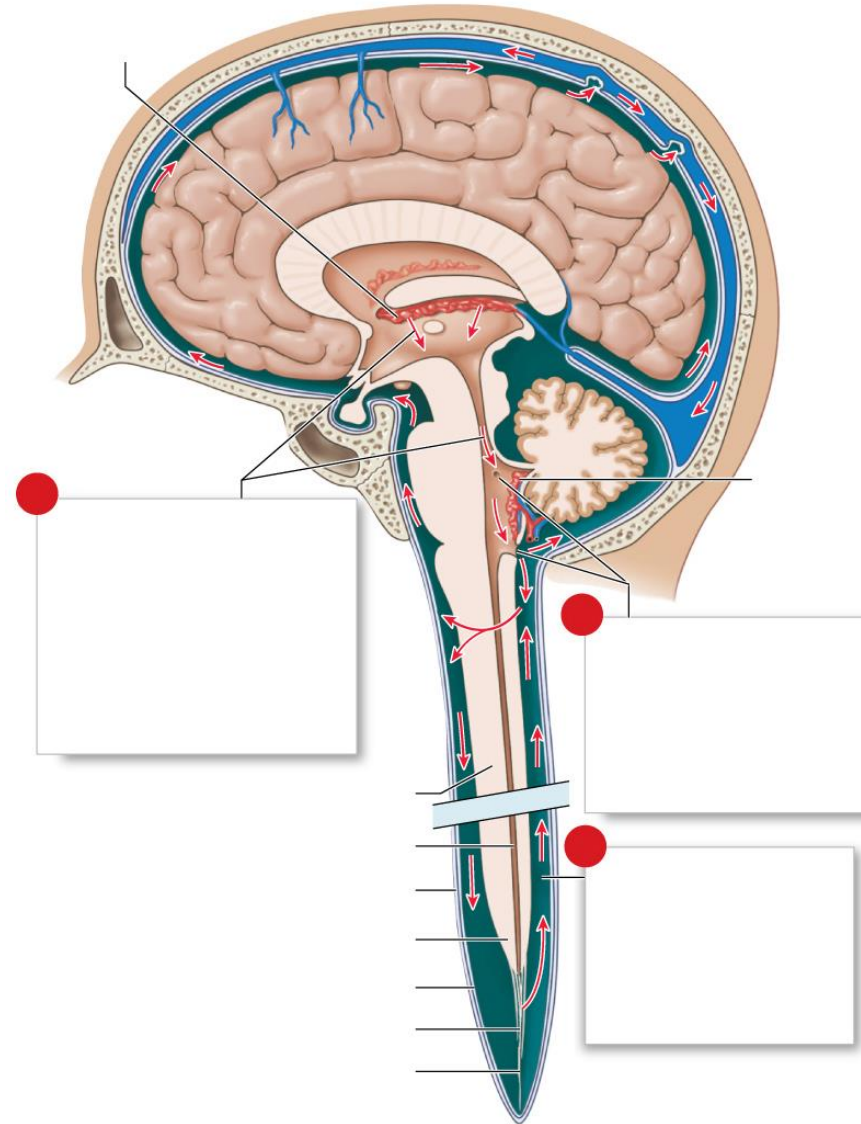
CSF Production & Circulation

- **Instructions**

- Work in small groups (~5 students)
- A patient has a blockage in the cerebral aqueduct.
- In your groups:
 - **Describe the production, flow path, and reabsorption of CSF**
 - **Predict the effect of the blockage on the ventricles and intracranial pressure**

- **Bloom's taxonomy = Apply/Analyze**

CSF Production & Circulation



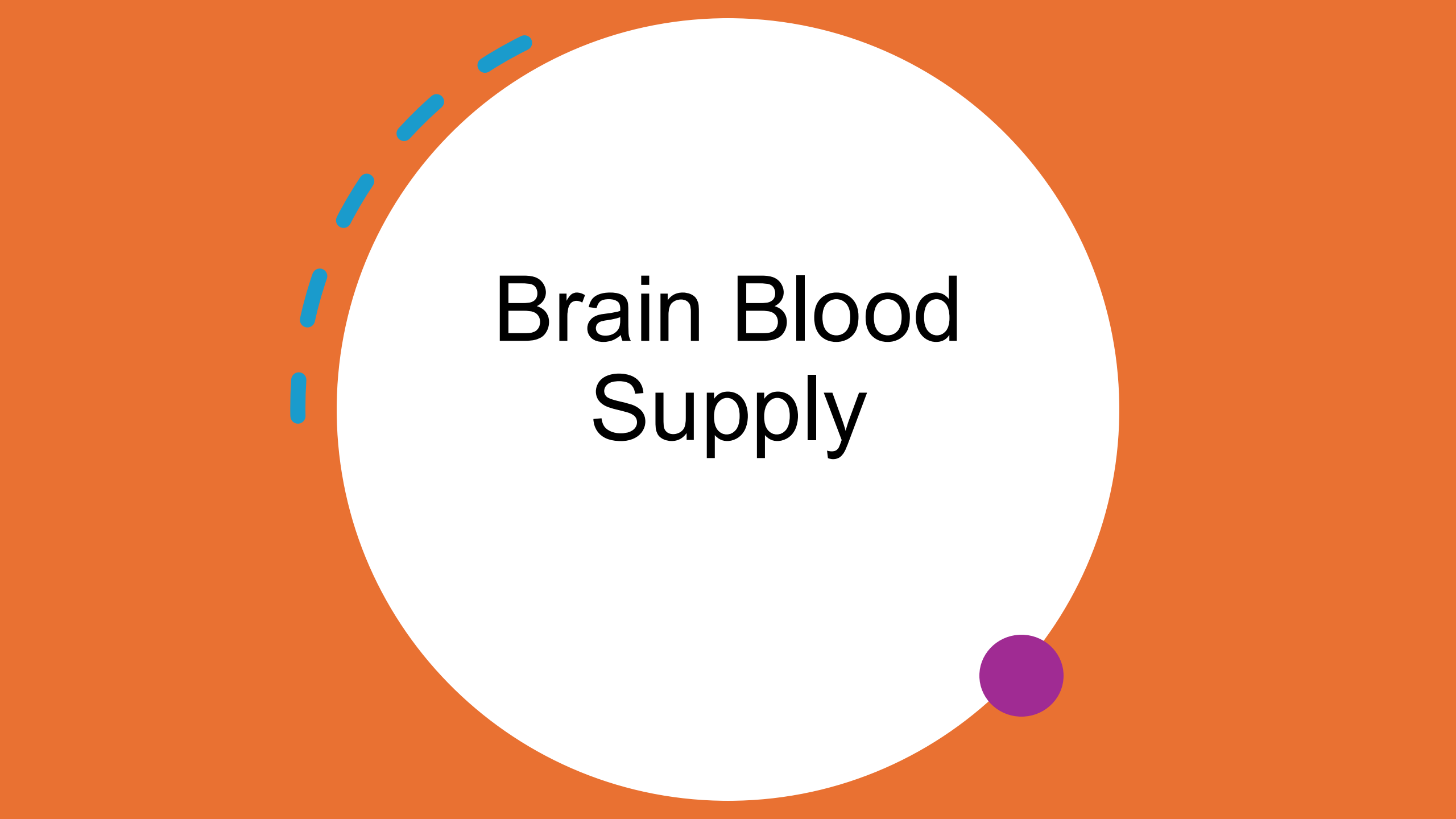
CSF Production & Circulation

- **Flow Path**

- Lateral ventricles → Interventricular foramina → Third ventricle → Cerebral aqueduct → Fourth ventricle → Subarachnoid space → Arachnoid granulations → Dural sinuses → Internal jugular vein

- **Predict the Effects**

- Blockage at the cerebral aqueduct causes CSF to accumulate in the lateral and third ventricles, leading to hydrocephalus.



Brain Blood Supply

Brain Blood Supply

- **Instructions**

- Work in pairs
- You will first **match each term to its correct description**
- You will then **use what you know to identify which component of the blood supply is most affected in this scenario**

A patient experiences paralysis on one side of the body and difficulty speaking

- **Bloom's taxonomy = Understand/Apply/Analyze**

Brain Blood Supply

Term	Description
1. Internal carotid arteries	A. Drain blood from the dural sinuses back to the heart
2. Vertebral arteries	B. Supply the anterior and middle cerebrum (e.g., frontal, parietal lobes)
3. Internal jugular veins	C. Supply the brainstem, cerebellum, and posterior cerebrum
4. Cerebrovascular accident (CVA)	D. Loss of brain function due to interruption of blood flow

Brain Blood Supply

Term	Match	Explanation
Internal carotid arteries	B	Carry oxygenated blood to anterior and middle brain regions (speech, motor control areas).
Vertebral arteries	C	Unite to form the basilar artery; supply brainstem, cerebellum, and occipital lobe.
Internal jugular veins	A	Major venous drainage pathway from dural sinuses to heart.
Cerebrovascular accident	D	Blockage (ischemic) or rupture (hemorrhagic) causes death to nervous tissue

Given the patient has trouble speaking, they likely have damage to the internal carotid arteries)



Brainstem & Diencephalon Sorting

Brainstem & Diencephalon Sorting

- **Instructions**

- Work in small groups (~5 students)
- **Match the structures to their correct structure**
- Structures form the 'core' of the brain, including many essential autonomic processes and relaying information
 - **Without these structures, we would not be able to function**
- **Bloom's taxonomy = Apply/Analyze/Evaluate**

Brainstem & Diencephalon Sorting

Structure	Function
Medulla oblongata	A. Visual reflex center
Pons	B. Maintains consciousness and alertness
Superior colliculus	C. Links cerebellum to brainstem; controls breathing rhythm
Inferior colliculus	D. Coordinates motor output and balance
Reticular activating system	E. Regulates homeostasis, temperature, hunger, and hormone secretion
Cerebellum	F. Autonomic control of HR, BP, and respiration
Thalamus	G. Auditory reflex center
Hypothalamus	H. Secretes melatonin; controls circadian rhythms
Pineal gland	I. Relays sensory info to cerebral cortex

Brainstem & Diencephalon Sorting

Structure	Match	Explanation
Medulla oblongata	F	Controls vital reflexes such as heartbeat, breathing, and vasomotor tone; essential for life.
Pons	C	Acts as a communication bridge and coordinates respiratory patterns.
Superior colliculus	A	Integrates visual stimuli to direct eye and head movements.
Inferior colliculus	G	Processes auditory input and initiates reflexive responses to sound.
Reticular activating system	B	Regulates arousal and awareness; damage causes coma.
Cerebellum	D	Ensures smooth, coordinated motor activity and equilibrium.
Thalamus	I	Processes and routes nearly all sensory input (except smell).
Hypothalamus	E	Maintains internal balance and controls the pituitary gland.
Pineal gland	H	Endocrine gland controlling biological rhythms through melatonin secretion.

Limbic System Integration



Limbic System Integration

- **Instructions**

- **You will be presented with different scenarios where you need to identify the limbic structure that is primarily affected**
 - Think on your own for 1 minute
 - Discuss with a partner for 1 minute
 - I will ask for volunteers to share their thinking and reasoning
- **Bloom's taxonomy = Evaluate/Create**

Limbic System Integration: Scenario 1

- A 48-year-old man survives a viral infection that damages part of his medial temporal lobe. He can hold a conversation normally, but if you leave the room and return five minutes later, he greets you as if for the first time. Interestingly, he can still play the piano pieces he learned years ago. **Which limbic structure is primarily affected?**
- **Structure affected:** Hippocampus
- **Explanation:** The hippocampus is critical for forming **new long-term declarative memories** (facts and events)
 - Damage leads to **anterograde amnesia** — the inability to form new memories

Limbic System Integration: Scenario 2

- A 36-year-old woman experienced cranial damage due to a rare genetic disorder. She reports that she no longer feels fear, even when confronted with dangerous situations. Friends note she seems emotionally “flat” and doesn’t react strongly to either good or bad news. **Which limbic structure is affected?**
- **Structure affected:** Amygdala
- **Explanation:** The amygdala generates **emotional responses** (especially fear and aggression) and assigns **emotional significance to experiences**

Limbic System Integration: Scenario 3

- A 72-year-old stroke survivor has no motor weakness or sensory loss, but her family reports she sits for hours doing nothing. When asked if she wants to eat, she replies calmly, “I suppose so,” but makes no effort to do so. She seems indifferent to her surroundings, though her speech and memory are intact. **Which limbic structure might be affected?**
- **Structure affected:** Cingulate gyrus
- **Explanation:** The cingulate gyrus connects emotional states to **motivated behavior** — it drives action based on internal emotional cues.